

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	S.PRATEEKSHA	Reg. No.:	192572137
Date:	21\08\26	Duration:	60 minutes

Code of Conduct

I S.PRATEEKSHA-192572137 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:S.PRATEEKSHA

Employee performance evaluations and productivity analytics system

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, design and document comprehensive test cases to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the functional and non-functional requirements to be tested.
2. Identify the test scenarios for the assigned problem statement.
3. Prepare detailed test cases covering normal, boundary, invalid, and exceptional conditions.
4. Include Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status.
5. Apply appropriate test design techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate requirement and functional coverage for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15 %)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25 %)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10 %)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

Annexure A – Test Case Design Assignment

Employee Performance Evaluations and Productivity Analytics System

1. Identify the functional and non-functional requirements to be tested.

1.1 Functional Requirements

ID	Functional Requirement
FR01	Users should be able to log in using valid credentials.
FR02	Authorized users should be able to add employee information.
FR03	Authorized users should be able to update employee information.
FR04	Users should be able to search and filter employee records.
FR05	Managers should be able to enter performance evaluation scores.
FR06	The system should calculate the overall performance score correctly.
FR07	The system should analyze employee productivity.
FR08	The system should classify employees as Excellent, Good, Average, or Poor.
FR09	The system should generate performance and productivity reports.
FR10	The system should validate empty, invalid, duplicate, and out-of-range inputs.
FR11	Only authorized users should access restricted employee performance functions.
FR12	Users should be able to log out securely.

1.2 Non-Functional Requirements

ID	Requirement	Testing Focus
NFR01	Usability	Interface should be easy to understand and navigate.
NFR02	Performance	Pages, calculations and reports should respond within reasonable time.
NFR03	Reliability	System should operate without unexpected crashes or inconsistent results.
NFR04	Availability	Application should be accessible when required services are running.
NFR05	Security	Unauthorized users and unsafe inputs should be handled safely.
NFR06	Maintainability	Modules and code structure should be understandable and modifiable.
NFR07	Scalability	System should support additional employees and evaluation records.
NFR08	Compatibility	Application should work correctly in supported browsers/devices.
NFR09	Data Integrity	Employee and evaluation information should remain accurate during operations.
NFR10	Accuracy	Scores, classifications and reports should be calculated correctly.

2. Identify the test scenarios for the assigned problem statement.

The following scenarios cover the major functions of the Employee Performance Evaluations and Productivity Analytics System.

- **TS01 – User Login**
Verify that authorized users can log in successfully.
- **TS02 – Invalid Login**
Verify that invalid credentials are rejected.
- **TS03 – Employee Addition**
Verify that valid employee information can be added.
- **TS04 – Employee Update**
Verify that existing employee information can be updated.
- **TS05 – Employee Search/Filtering**
Verify that users can find employees using search/filter options.
- **TS06 – Performance Evaluation**
Verify that managers can enter valid evaluation scores.
- **TS07 – Score Calculation**
Verify that overall performance scores are calculated correctly.
- **TS08 – Productivity Analysis**
Verify that productivity metrics are analyzed correctly.
- **TS09 – Performance Classification**
Verify score-based classification.
- **TS10 – Invalid Input**
Verify that empty, non-numeric and out-of-range data is rejected.
- **TS11 – Report Generation**
Verify that performance/productivity reports are generated.
- **TS12 – Authorization & Logout**
Verify restricted access and secure logout.
- **TS13 – Error Handling**
Verify appropriate handling of unavailable or invalid requests.
- **TS14 – Performance & Reliability**
Verify stable operation with repeated and larger data operations.

Figure: Proposed system architecture



3. Prepare detailed test cases covering normal, boundary, invalid, and exceptional conditions.

3.1 Equivalence Partitioning

Inputs are divided into valid and invalid groups. Example: overall performance score must be between 0 and 100.

Partition	Example	Expected
Valid	0–100	Accept
Invalid (Low)	-1 or less	Reject
Invalid (High)	101 or more	Reject
Invalid (Non-numeric)	ABC	Reject
Empty	Blank	Reject

3.2 Boundary Value Analysis

Test values around the minimum and maximum allowed score.

Test Value	Expected Result
-1	Reject
0	Accept
1	Accept
99	Accept
100	Accept
101	Reject

3.3 Decision Table

Score Range	Condition	Classification
90–100	$\text{Score} \geq 90$	Excellent
75–89	$75 \leq \text{Score} < 90$	Good
50–74	$50 \leq \text{Score} < 75$	Average
0–49	$\text{Score} < 50$	Poor

3.4 State Transition Testing



Figure: Employee evaluation workflow

4. Detailed Test Cases

Test cases include normal, boundary, invalid, security, performance, and exceptional conditions.

TC01 – Valid User Login

Test Case ID	TC01
Test Scenario	Valid User Login
Test Steps	1. Open login page 2. Enter username 3. Enter password 4. Click Login
Test Data	Username: manager01; Password: valid@123
Expected Result	Dashboard should open and user should be authenticated.
Actual Result	Not executed
Status	N/A

TC02 – Invalid User Login

Test Case ID	TC02
Test Scenario	Invalid User Login
Test Steps	1. Open login page 2. Enter incorrect credentials 3. Click Login
Test Data	Username: manager01; Password: wrong123
Expected Result	System should reject login and display an error message.
Actual Result	Not executed
Status	N/A

TC03 – Add Employee

Test Case ID	TC03
Test Scenario	Add Employee
Test Steps	1. Open employee module 2. Click Add Employee 3. Enter details 4. Save
Test Data	ID: EMP1001; Name: Ravi Kumar; Dept: IT
Expected Result	Employee record should be created successfully.
Actual Result	Not executed
Status	N/A

TC04 – Duplicate Employee ID

Test Case ID	TC04
Test Scenario	Duplicate Employee ID
Test Steps	1. Open Add Employee 2. Enter an existing ID 3. Save
Test Data	Employee ID: EMP1001
Expected Result	System should reject duplicate ID and display validation message.
Actual Result	Not executed
Status	N/A

Employee Performance Dashboard

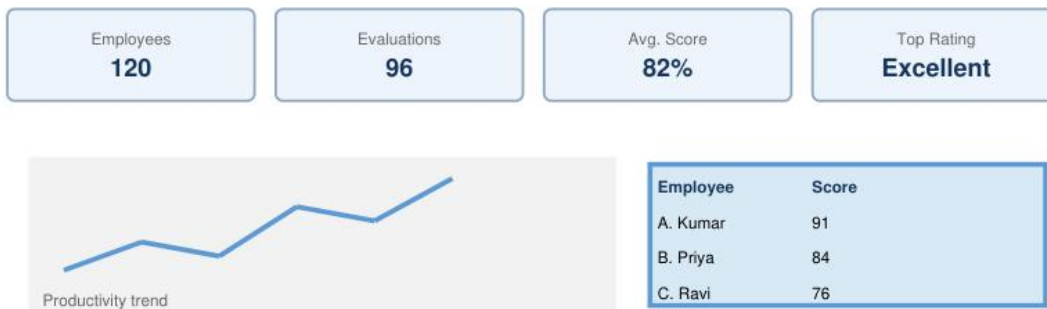


Figure: Illustrative employee performance dashboard

Figure 1: Illustrative employee performance dashboard for the proposed system.

4. Detailed Test Cases

Test cases include normal, boundary, invalid, security, performance, and exceptional conditions.

TC05 – Update Employee

Test Case ID	TC05
Test Scenario	Update Employee
Test Steps	1. Search employee 2. Edit department 3. Save
Test Data	EMP1001; Dept: CSE
Expected Result	Updated employee information should be stored correctly.
Actual Result	Not executed
Status	N/A

TC06 – Search Employee

Test Case ID	TC06
Test Scenario	Search Employee
Test Steps	1. Open employee list 2. Enter employee ID 3. Search
Test Data	EMP1001
Expected Result	Correct employee record should be displayed.
Actual Result	Not executed
Status	N/A

TC07 – Valid Performance Scores

Test Case ID	TC07
Test Scenario	Valid Performance Scores
Test Steps	1. Open evaluation page 2. Enter valid scores 3. Submit
Test Data	Attendance 90; Quality 85; Teamwork 80; Punctuality 95
Expected Result	All values should be accepted.
Actual Result	Not executed
Status	N/A

TC08 – Minimum Score Boundary

Test Case ID	TC08
Test Scenario	Minimum Score Boundary
Test Steps	1. Open evaluation 2. Enter minimum score 3. Submit
Test Data	All scores = 0
Expected Result	Zero should be accepted as the minimum valid value.
Actual Result	Not executed
Status	N/A

4. Detailed Test Cases

Test cases include normal, boundary, invalid, security, performance, and exceptional conditions.

TC09 – Maximum Score Boundary

Test Case ID	TC09
Test Scenario	Maximum Score Boundary
Test Steps	1. Open evaluation 2. Enter maximum score 3. Submit
Test Data	All scores = 100
Expected Result	100 should be accepted as the maximum valid value.
Actual Result	Not executed
Status	N/A

TC10 – Score Below Minimum

Test Case ID	TC10
Test Scenario	Score Below Minimum
Test Steps	1. Enter a score below 0 2. Submit
Test Data	Score = -1
Expected Result	System should reject the value.
Actual Result	Not executed
Status	N/A

TC11 – Score Above Maximum

Test Case ID	TC11
Test Scenario	Score Above Maximum
Test Steps	1. Enter a score above 100 2. Submit
Test Data	Score = 101
Expected Result	System should reject the value.
Actual Result	Not executed
Status	N/A

TC12 – Non-Numeric Score

Test Case ID	TC12
Test Scenario	Non-Numeric Score
Test Steps	1. Enter text in a score field 2. Submit
Test Data	Attendance = ABC
Expected Result	System should reject non-numeric input and show validation.
Actual Result	Not executed
Status	N/A

Performance Classification

90–100 → Excellent | 75–89 → Good
50–74 → Average | 0–49 → Poor

Figure 2: Performance classification illustration.

4. Detailed Test Cases

Test cases include normal, boundary, invalid, security, performance, and exceptional conditions.

TC13 – Overall Score Calculation

Test Case ID	TC13
Test Scenario	Overall Score Calculation
Test Steps	1. Enter valid scores 2. Click Calculate
Test Data	90, 80, 70, 100, 80
Expected Result	System should calculate the correct overall score based on the defined form...
Actual Result	Not executed
Status	N/A

TC14 – Performance Classification

Test Case ID	TC14
Test Scenario	Performance Classification
Test Steps	1. Calculate overall score 2. View classification
Test Data	Overall score = 92
Expected Result	Employee should be classified as Excellent.
Actual Result	Not executed
Status	N/A

TC15 – Productivity Analysis

Test Case ID	TC15
Test Scenario	Productivity Analysis
Test Steps	1. Select employee 2. Open analytics 3. Generate analysis
Test Data	Tasks completed: 45; Target: 50; Attendance: 95%
Expected Result	Productivity result and relevant metrics should be displayed.
Actual Result	Not executed
Status	N/A

TC16 – Generate Performance Report

Test Case ID	TC16
Test Scenario	Generate Performance Report
Test Steps	1. Select employee 2. Click Generate Report
Test Data	EMP1001; Monthly report
Expected Result	Report should be generated with correct employee and performance details.
Actual Result	Not executed
Status	N/A

4. Detailed Test Cases

Test cases include normal, boundary, invalid, security, performance, and exceptional conditions.

TC17 – Filter by Classification

Test Case ID	TC17
Test Scenario	Filter by Classification
Test Steps	1. Open employee analytics 2. Select Excellent 3. Apply filter
Test Data	Classification = Excellent
Expected Result	Only employees in the selected classification should be displayed.
Actual Result	Not executed
Status	N/A

TC18 – Unauthorized Access

Test Case ID	TC18
Test Scenario	Unauthorized Access
Test Steps	1. Log in as restricted user 2. Open manager evaluation page
Test Data	Role = Employee
Expected Result	Restricted manager functions should not be accessible.
Actual Result	Not executed
Status	N/A

TC19 – Logout and Session Check

Test Case ID	TC19
Test Scenario	Logout and Session Check
Test Steps	1. Click Logout 2. Try opening protected page
Test Data	Active session after logout
Expected Result	User should be logged out and protected page should require authentication.
Actual Result	Not executed
Status	N/A

TC20 – Large Dataset / Reliability

Test Case ID	TC20
Test Scenario	Large Dataset / Reliability
Test Steps	1. Load many employee records 2. Search 3. Generate report repeatedly
Test Data	1000+ employee records
Expected Result	System should remain stable and produce correct results without unexpected ...
Actual Result	Not executed
Status	N/A



Figure: Employee evaluation workflow

5. Requirement Traceability and Test Coverage

Requirement	Test Cases Covered	Coverage
FR01 – User Login	TC01, TC02	✓
FR02 – Employee Addition	TC03, TC04	✓
FR03 – Employee Update	TC05	✓
FR04 – Search/Filtering	TC06, TC17	✓
FR05 – Performance Evaluation	TC07–TC12	✓
FR06 – Score Calculation	TC13	✓
FR07 – Productivity Analysis	TC15	✓
FR08 – Performance Classification	TC14	✓
FR09 – Report Generation	TC16	✓
FR10 – Data Validation	TC04, TC08–TC12	✓
FR11 – Authorization	TC18	✓
FR12 – Logout	TC19	✓
NFR01 – Usability	TC01, TC06, TC16	✓
NFR02 – Performance	TC20	✓
NFR03 – Reliability	TC19, TC20	✓
NFR05 – Security	TC02, TC18, TC19	✓
NFR09 – Data Integrity	TC03–TC05, TC13, TC16	✓
NFR10 – Accuracy	TC13–TC16	✓

6. Test Execution Record

The following fields should be updated after executing the implemented system. The current document records the designed expected results; Actual Result and Pass/Fail are intentionally marked as pending to avoid claiming unexecuted tests as completed.

Metric	Result
Total Designed Test Cases	20
Functional Test Cases	14
Non-Functional Test Cases	6
Boundary Value Tests	4
Equivalence Partition Tests	5
Decision Table Tests	4
State Transition Tests	6
Requirement Coverage	100% of listed requirements mapped

7. Ensure that the test cases provide adequate requirement and functional coverage for the assigned system.

Test Coverage Summary

Metric	Result
Total Test Cases	20
Functional Tests	14
Non-Functional Tests	6
Boundary Tests	4
Equivalence Partition Tests	5
Decision Table Tests	4
State Transition Tests	6
Security Tests	4
Validation Tests	8
Requirement Coverage	100% of identified requirements

Key Coverage Points

- ✓ Functional testing covers login, employee management, evaluation, calculation, analytics, classification, reports, search and logout.
- ✓ Validation testing covers empty, duplicate, non-numeric, negative and out-of-range inputs.
- ✓ Boundary Value Analysis covers the minimum and maximum performance-score limits.
- ✓ Equivalence Partitioning divides score inputs into valid and invalid classes.
- ✓ Decision Table Testing verifies score-based performance classification.
- ✓ State Transition Testing verifies the employee evaluation lifecycle.
- ✓ Security testing covers authentication, authorization and session termination.
- ✓ Performance and reliability testing considers larger datasets and repeated operations.

Figure: Proposed system architecture

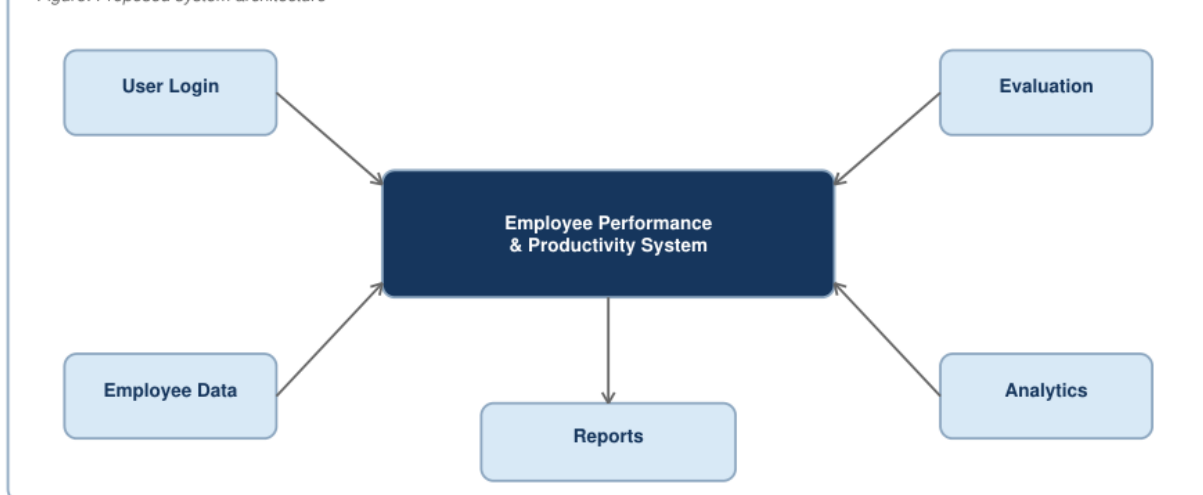


Figure 3: Overall system architecture supporting employee performance evaluation and productivity analytics.

Conclusion

The designed test cases provide structured coverage of the Employee Performance Evaluations and Productivity Analytics System. Normal, boundary, invalid, security, performance, and exceptional conditions are included. The use of Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing improves the completeness and traceability of the testing process. Actual Result and Pass/Fail fields can be updated after execution of the implemented system.